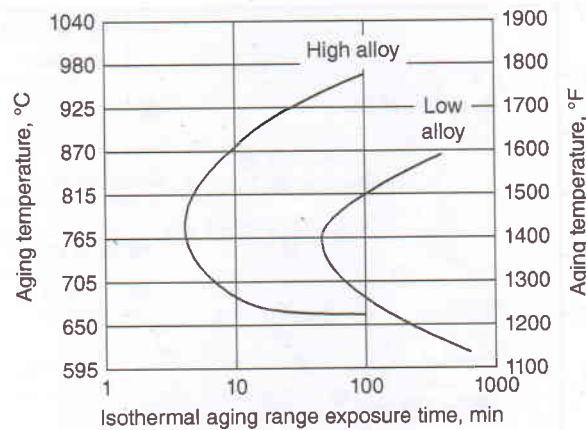




**Figure 12.10** Nickel-based alloys with reduced amounts of aluminum+titanium result in the nose of the precipitation reaction curve moving to much longer times (Source: Berry and Hughes, *Welding Journal*, vol.46)

imposes a hold time below the nose of the precipitation curve prior to heating to the solution temperature. This will provide some reduction in residual stresses in order to mitigate the stress relaxation that coincides with the precipitation reaction.

Since nickel-based alloys have fully austenitic microstructures, weld solidification cracking is a potential problem under high restraint conditions. HAZ liquation cracking can also occur, particularly in some of the superalloys. In most cases, solidification and liquation cracking susceptibility can be reduced by minimizing impurity concentrations as well as restraint.

### 12.3 Titanium Alloys

Titanium alloys exhibit low density and can be strengthened to high levels by a combination of solid solution and transformation hardening. As a result, they are known for their superior strength-to-weight ratio, or specific strength. In addition, they offer excellent corrosion resistance in most environments, including seawater. Some of the alloys can be used at temperatures as high as 540°C (1000°F). Titanium alloys are very expensive, primarily due to the very time consuming and expensive chemical extraction process used to separate titanium from its ore (rutile).

There are four general classes of titanium alloys, which are based on their microstructure — commercially pure (CP), alpha, alpha-beta, and metastable beta alloys. Like iron, titanium exhibits allotropic behavior, which allows for transformation hardening. At low temperatures, the crystal structure is hexagonal close packed (HCP), and at elevated temperatures it transforms to BCC. The HCP phase is known as alpha and the BCC phase is known as beta. By controlling the proportion of alpha-stabilizing (examples are aluminum, tin, and zirconium) to beta-stabilizing elements (examples are vanadium, molybdenum, and chromium), the beta phase can remain stable at room temperature, and mixtures of these phases are possible. Hard martensitic microstructures can form under rapid cooling conditions, often requiring postweld heat treatments. However, the weldability concern associated with martensite formation is minimal when compared to that associated with steels.

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**Figure 1**

**Figure 12**

Titanium alloys are usually selected for either their corrosion resistance or specific strength. Applications that rely on their corrosion resistance include heat exchangers, pressure vessels, waste storage, and tube and piping. They are resistant to stress corrosion cracking and perform well in marine environments. They are also biocompatible, which makes them a good choice for medical implants (Figure 12.11) such as hips, knees, and fasteners.

The widespread use of titanium alloys in the aerospace industry is driven almost exclusively by their high specific strength. In particular, high-performance military aircraft (Figure 12.12)



**Figure 12.11** Medical devices and implants are often made out of titanium (Source: Zimmer)



**Figure 12.12** Titanium is used extensively in many military aircraft (Source: Lockheed Martin)



**Figure 12.13** Colored oxide scale and/or alpha case formation is indicative of improper shielding when welding titanium

rely heavily on titanium alloys for both air frames and skins. Their high specific strength can also create a performance advantage when they are used in sports equipment, such as bicycle frames, tennis rackets, and golf clubs.

Titanium alloys are weldable with most processes (except Shielded Metal Arc Welding (SMAW)) if proper precautions are taken. Two of the biggest welding concerns are interstitial embrittlement (or contamination cracking) and excessive HAZ and weld metal grain size. Titanium readily absorbs and dissolves interstitial elements such as oxygen, nitrogen, and hydrogen. In small amounts, these elements strengthen titanium quite profoundly. For example, the difference in tensile strengths among the CP titanium grades is mainly due to slight variations in oxygen and nitrogen content. However, at larger amounts of absorption, significant embrittlement occurs. Embrittlement can occur quite rapidly above 500°C (930°F), which means that during welding, even regions on the part that are relatively far from the fusion zone are susceptible. Since the ambient atmosphere contains all of the embrittling elements, protection (shielding) during welding is especially important. Therefore, when welding titanium it is common to use more thorough methods of shielding such as the use of larger nozzles, trailing shields, back-side shielding or purging, and in some cases welding in a glove box purged with an inert gas such as argon. It is also very important to properly remove oxides and contaminants prior to welding, and also to use gases of high purity.

The degree of contamination is often determined by the color of the oxide that forms during welding (Figure 12.13). The oxide color is a function of its thickness. A light silver or bronze-colored oxide indicates a mild contamination, while blue or purple oxides are an indication of more extensive atmospheric contamination and may result in the rejection of the weld based on the suspicion of embrittlement. A white, flaky oxide indicates severe contamination. However, it is important to remember that the oxide can be easily removed by wire brushing following welding; so complete reliance on the oxide color as a measurement of weld quality is not always the best approach.

A layer known as alpha case (Figure 12.13) may also form if shielding is not sufficient. Alpha case is a layer of titanium that has been highly enriched in alpha-stabilizing elements such as oxygen and nitrogen. It is brittle and susceptible to forming small cracks that can significantly reduce mechanical properties such as fatigue.

Titanium alloys solidify as a single phase BCC beta structure and remain that way for some time during cooling. Since there are no other phases present to retard grain growth, weld metal and HAZ grain sizes can become quite large, resulting in significant reductions in ductility. The best approach to minimizing grain sizes is by reducing weld heat input. In particular, extremely low heat input welding processes such as Electron Beam and Laser Welding can help keep grain sizes relatively small. Electron Beam Welding can be especially effective

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since welding in a vacuum essentially eliminates any risk of interstitial embrittlement. In the right application, Diffusion Welding is also an effective method for welding titanium. This is because titanium readily dissolves oxides and exhibits low creep strengths at elevated temperatures, two traits that are ideal for the Diffusion Welding process.

## 12.4 Copper Alloys

Copper alloys are used in a variety of engineering applications, but are selected typically because of their electrical and thermal conductivity as well as corrosion resistance. With the exception of alloys containing beryllium (strengthened by precipitation hardening), copper alloys are strengthened by solid solution and cold work, and in general have low to moderate strength but good ductility and toughness. Their microstructures are either single phase FCC or dual phase FCC+BCC. The family of copper alloys includes brass, a copper alloy containing zinc and possibly some tin or lead, and bronze, which is copper alloyed with tin, aluminum, or silicon.

Their resistance to salt water corrosion makes them a good choice for marine applications such as tubing, boilers, and decorative hardware on boats and ships. Other applications that make use of their excellent corrosion resistance range from storage containers for nuclear fuel to caskets. They can be easily fabricated into a variety of shapes and can be readily cast. For example, early cannons were produced from copper alloy castings.

Copper alloys are also used for a variety of architectural and artistic purposes. The copper oxide that forms in the atmosphere can take on a variety of shades (depending on alloying elements added) from gold to red to brown, and thus is very versatile for creating different artistic effects. Under normal atmospheric exposure many of the Cu alloys will develop a greenish oxide, such as that seen over time on copper-roofed buildings. The physical properties of these alloys also make them popular in the musical instrument market, and for the production of bells and chimes. Most of the famous bells worldwide are cast from brass or bronze alloys.

Since cold work is a primary method for strengthening copper alloys, welding will result in HAZ recrystallization and grain growth, creating a much softer region than the surrounding base metal. In heavily cold worked alloys, this loss of strength can be significant and will severely limit the load-bearing capacity of the structure. Generally, the best approach to minimizing HAZ softening is to reduce heat input, but the high thermal conductivity and diffusivity of copper results in such rapid heat extraction during welding that high heat input is usually necessary. In fact, in order to achieve sufficient penetration with many alloys, weld preheating often becomes a requirement, especially as part thicknesses increase above about 0.25 in. (Figure 12.14). The use of helium-shielding gas can also significantly improve weld penetration. An example of the effect of preheat and shielding gas type on depth of fusion is shown in Figure 12.15.

Alloys that contain Zn, Cd, or P may be susceptible to porosity formation. This can usually be controlled by using a porosity resistant filler metal, but can be problematic in autogenous welds in susceptible alloys. Removal of the oxide prior to welding by chemical or mechanical means is recommended. Most copper alloys are considered resistant to solidification and liquation cracking, even though they solidify as a single phase FCC structure. Alloys containing tin and nickel may be susceptible to cracking under high restraint conditions. Both of these elements tend to widen the solidification temperature range. Elements such as selenium, sulfur, and lead are often added to improve machinability, but also increase susceptibility to solidification cracking, and therefore, should be avoided for applications that involve welding.